TOPIC 7

Bias-Variance Tradeoff: Explained

The original class slides, with a plain explanation under each one

> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

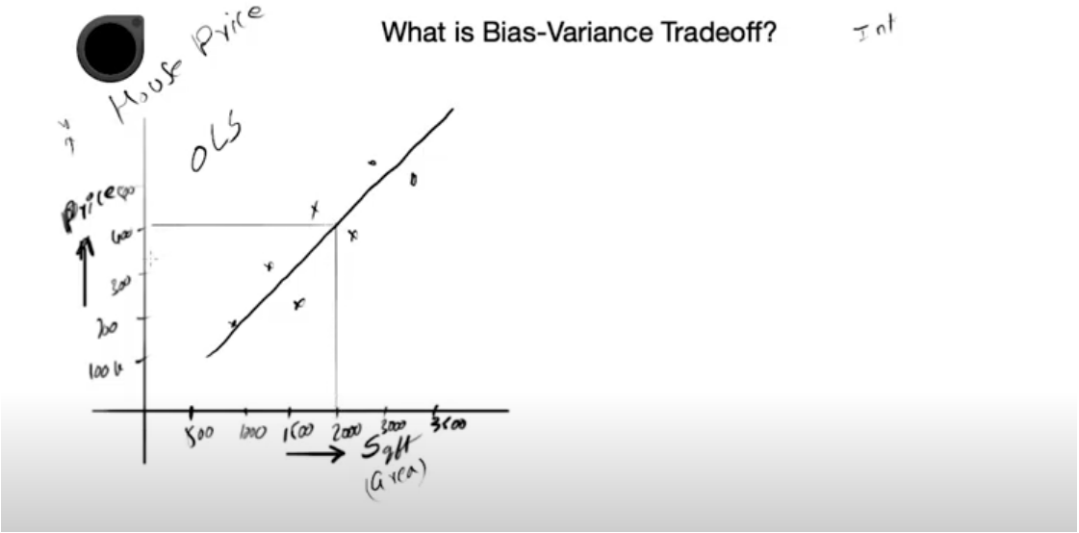
Bias and **variance** are the two kinds of error a model can make, and they pull in opposite directions. These 16 slides build the idea on a **house price against size (square feet)** whiteboard, then summarise it with definitions and a chart.

The key picture: a simple model (a line) misses the pattern but is stable; a very flexible model (a wiggly curve) matches the training houses perfectly but gets new houses wrong. The best model sits in between.

Words you will meet

Word	Plain meaning
Train data (blue)	Houses the model learns from
Test data (red)	New houses used to check the model
Bias	How far the model's average prediction is from the truth: error from being too simple
Variance	How much the model's predictions change with different training data: error from being too flexible
Underfitting	High bias: the model is too simple
Overfitting	High variance: the model chases the noise
Irreducible error	Noise in the data that no model can remove
Total error	Bias^2 + Variance + Irreducible error

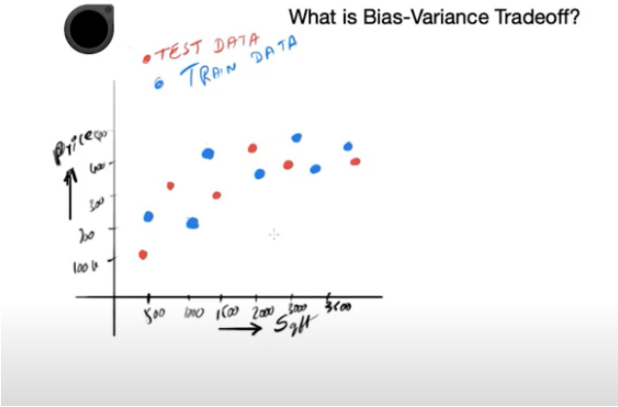
Slide 1: Starting question: price from size



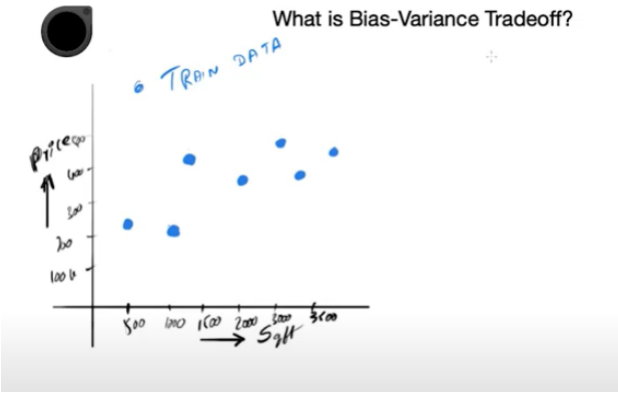
Slide 1

The title is **What is Bias-Variance Tradeoff?**. The sketch shows a line fitted by **OLS** (ordinary least squares, the standard line-fitting method) to predict **house price** from **size in square feet**. Reading across and down from the line gives the predicted price for a given size.

Slides 2 to 3: Two sets of houses: train and test



Slide 2

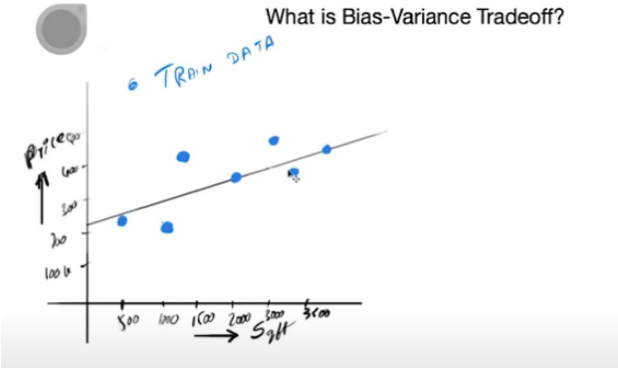


Slide 3

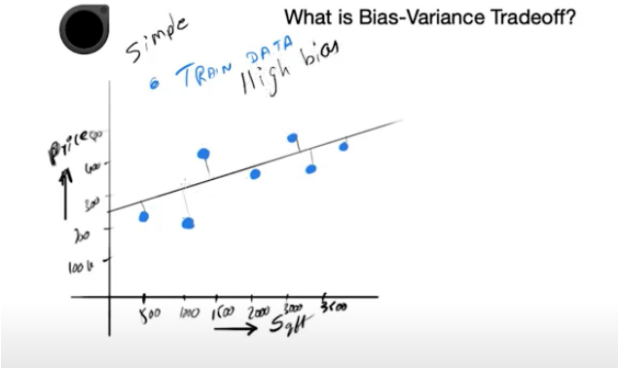
The dots are houses. **Blue dots are training data**: the houses the model is allowed to learn from. **Red dots are test data**: houses kept hidden to check how the model does on new houses. Slide 3 shows only the blue training dots.

- Price goes up the vertical axis, size (sqft) along the horizontal axis

Slides 4 to 5: A simple model: a straight line



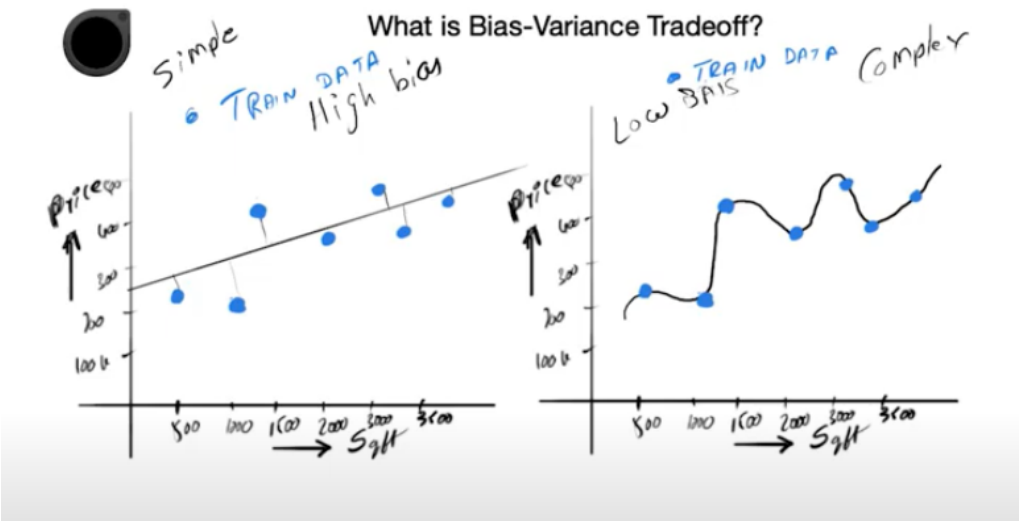
Slide 4



Slide 5

A straight line is fitted through the blue training dots. It is a **simple** model. It cannot bend, so it misses several dots (the short vertical lines on slide 5 show the gaps). Because the model is too simple for the pattern, it has **high bias**.

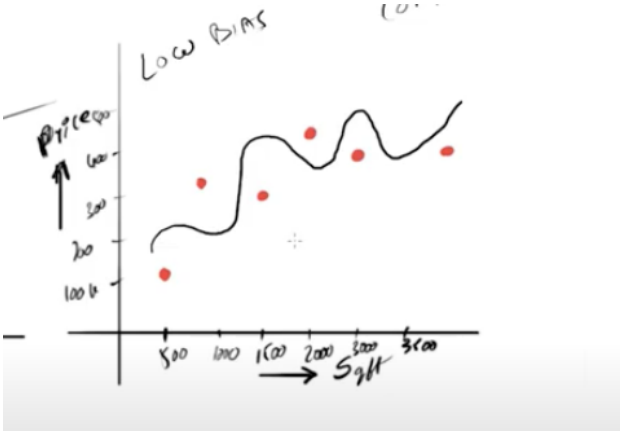
Slide 6: A complex model beside it



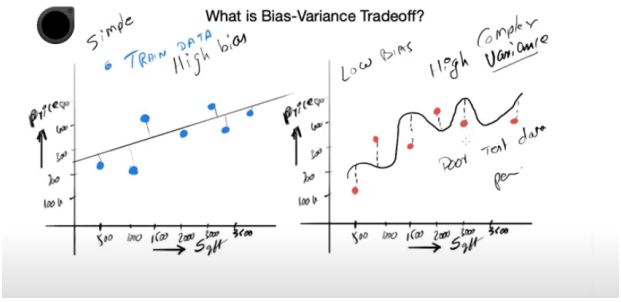
Slide 6

On the right a second, **complex** model is drawn: a wiggly curve that passes through (almost) every blue training dot. Its gaps to the training dots are tiny, so it has **low bias**. The simple line on the left has high bias.

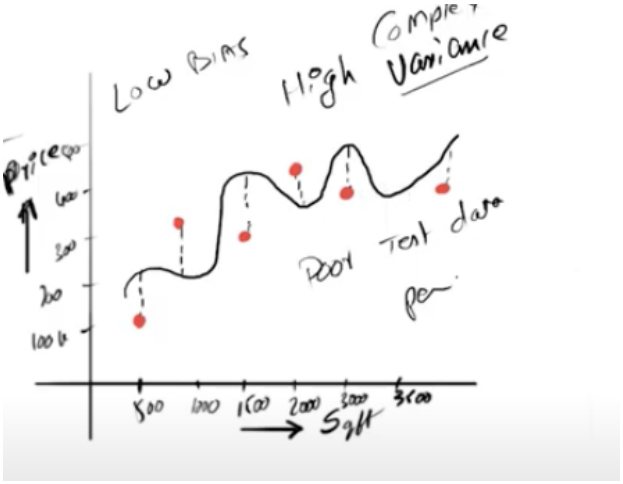
Slides 7 to 9: The complex model on new data



Slide 7



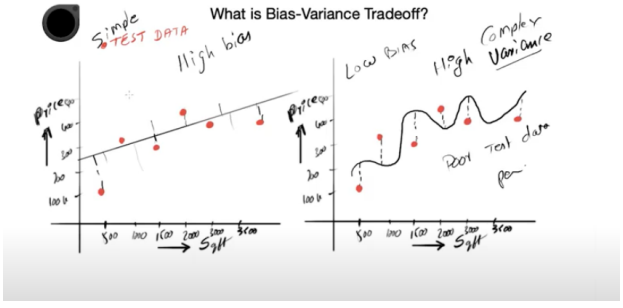
Slide 8



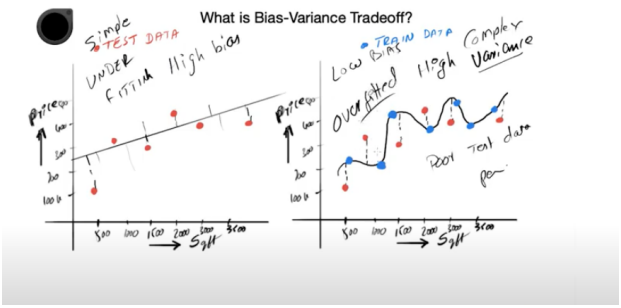
Slide 9

Now the red **test** houses are placed on the complex curve. The curve was shaped to fit the training dots exactly, so it lands far from the new red dots: the dotted vertical lines are the big errors. The board writes **High Variance** and **Poor Test data performance**. A model that is that sensitive to its training data gives very different curves for different samples, and does badly on new data.

Slides 10 to 11: Both models side by side on test data



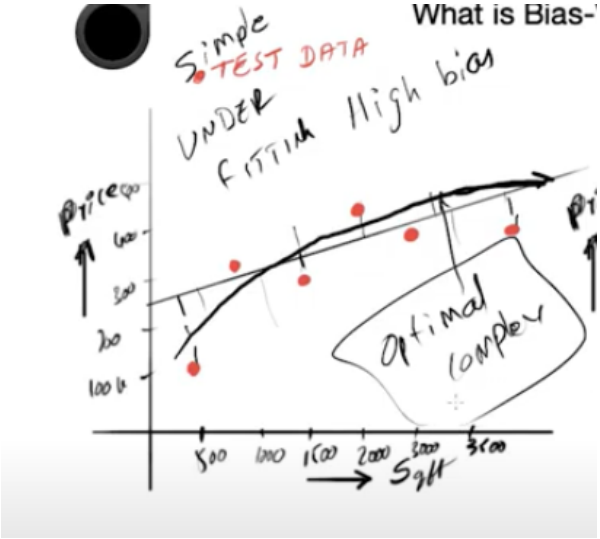
Slide 10



Slide 11

Left: the simple line with the test dots: it is **underfit** (it misses the pattern). Right: the complex curve with the test dots: it is **overfit** (it fits the training noise). Slide 11 labels them *under fitting, high bias* and *over fitted, low bias, high variance*.

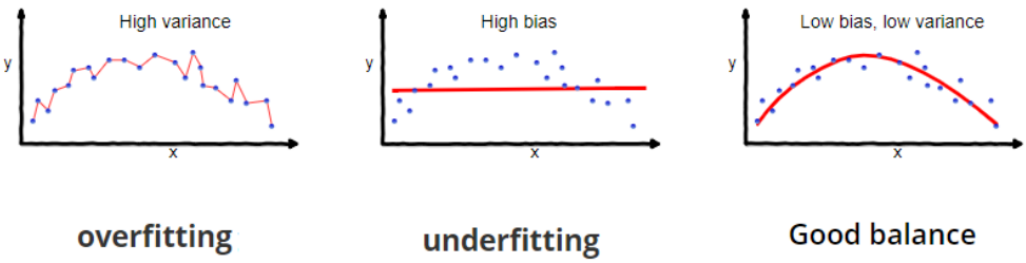
Slide 12: The goal: optimal complexity



Slide 12

The ideal model sits in between. It is flexible enough to follow the real trend (not too simple) and not so flexible that it chases the noise (not too complex). The board labels this **Optimal complexity**. It does reasonably well on the training data **and** on the test data.

Slide 13: Three pictures to remember

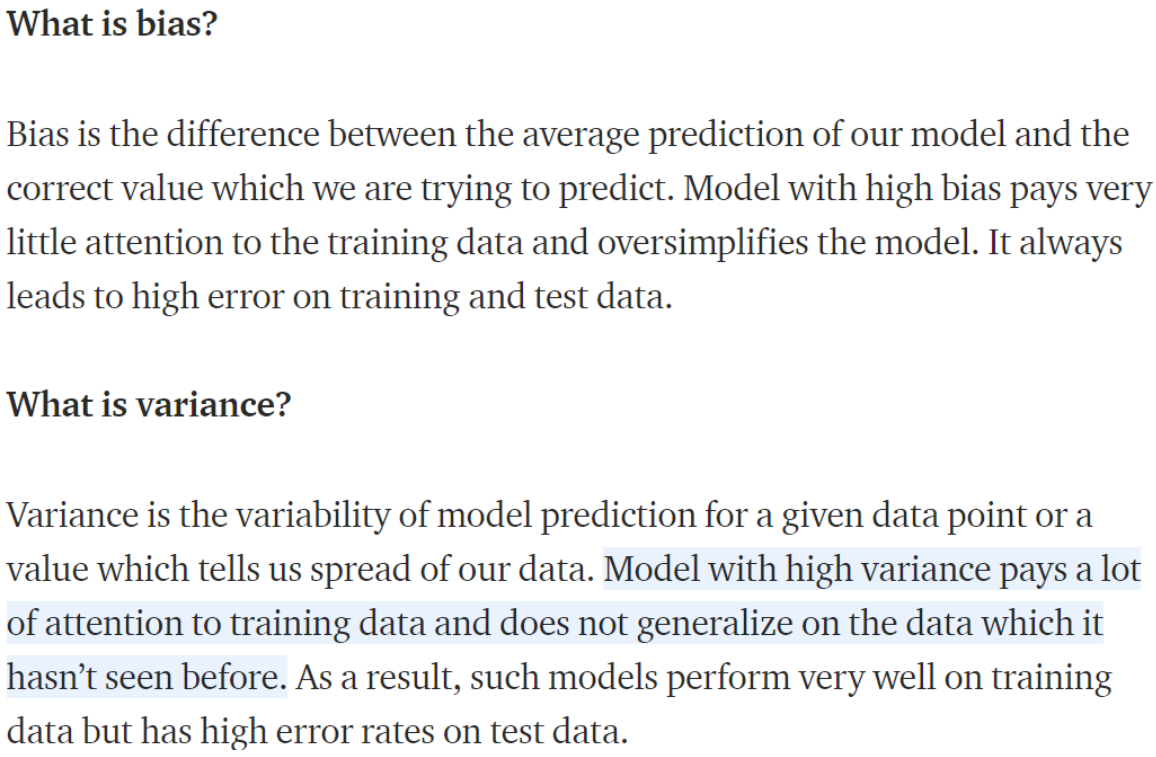


Slide 13

Three small graphs:

- **Overfitting (high variance):** a jagged line through every dot
- **Underfitting (high bias):** a flat line that ignores the pattern
- **Good balance (low bias, low variance):** a smooth curve that follows the trend

Slide 14: Definitions: bias and variance



Slide 14

Bias is the difference between the average prediction of the model and the correct value we are trying to predict. A model with **high bias** pays very little attention to the training data and over-simplifies; it always has **high error on training and test data**. **Variance** is how much the model's prediction varies for a given data point. A model with **high variance** pays a lot of attention to the training data and does not generalise to data it has not seen before: it performs very well on training data but has high error on test data.

Slide 15: Why it is a tradeoff

Why is Bias Variance Tradeoff?

If our model is too simple and has very few parameters then it may have high bias and low variance. On the other hand if our model has large number of parameters then it's going to have high variance and low bias. So we need to find the right/good balance without overfitting and underfitting the data.

This tradeoff in complexity is why there is a tradeoff between bias and variance. An algorithm can't be more complex and less complex at the same time.

Total Error

To build a good model, we need to find a good balance between bias and variance such that it minimizes the total error.

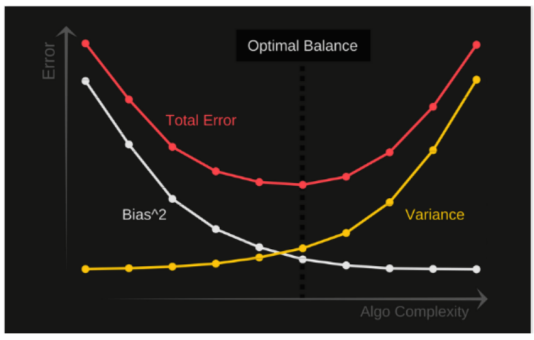
Total Error = Bias^2 + Variance + Irreducible Error

Slide 15

If the model is too simple (very few parameters) it may have **high bias and low variance**. If it has a large number of parameters, it will have **high variance and low bias**. We cannot have both at once, so we need the right balance, without overfitting or underfitting. This is the **tradeoff**. To build a good model we want the balance that **minimises the total error**:

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Slide 16: The chart: where the total error is lowest



An optimal balance of bias and variance would never overfit or underfit the model.

Therefore understanding bias and variance is critical for understanding the behavior of prediction models.

Slide 16

The horizontal axis is **algorithm complexity**; the vertical axis is error. The white curve (**Bias²**) falls as complexity grows. The yellow curve (**Variance**) rises. The red curve (**Total Error**) is their combination and has a lowest point: the **Optimal Balance**. An optimal balance would never overfit or underfit. Understanding bias and variance is critical to understanding how prediction models behave.